Subject: Robot Mechanics, Kinematics & Dynamics (OE-5)								
Program: B. Tech. (All Branches; Except Mech.)				Subject Code: ME0425			Semester: IV	
Teaching Scheme				Examination Evaluation Scheme				
Lecture	Tutorial	Practical	Credits	University Theory Examination	University Practical Examination	Continuous Internal Evaluation (CIE) Theory	Continuous Internal Evaluation (CIE) Practical	Total Marks
3	0	0	3	16/40	NA	24/60	NA	100

Course objectives:

This course aims to familiarize students with basic terminologies of the robotics and essential knowledge to be acquainted in the field of Robotics. It also aims to inculcate thorough understanding about basic terminologies, grippers, sensors, actuators and robot kinematics.

Course Outcomes (CO):

1. Understand relationship between joint space and Cartesian space
2. Able to calculate force and torque requirements for robotic joints
3. Plan trajectory of robot to reach the destination with controlled environment
4. Acquainted with methods of programing and various languages used in robotics
5. Awareness of AI and Machine learning in robotics

Course Contents:

Unit	Contents	Hours
1	**Velocities & Statics** Cross Product Operator for kinematics, Jacobians - Direct Differentiation, Basic Jacobian, JacobianJv / Jw, Jacobian in a Frame, Jacobian in Frame {0}, Kinematic Singularity.	6
2	**Robot Kinematics** Manipulator Kinematics, Link Description, Link to reference frame	7

	connections, Denavit-Hartenberg Approach, D-H Parameters, Position Representations, Forward Kinematics, Inverse Kinematics.	
3	**Robot Dynamics** Introduction to Dynamics, Lagrangian formulation of manipulator dynamics, dynamic simulation, computational consideration.	**4**
4	**Motion Control** Path versus Trajectory, Joint space versus Cartesian space descriptions, Point to Point Control, trajectory generation, Continuous Path Control, Force Control, hybrid position/force control system.	**6**

Reference Books:

1. S. K. Saha, Introduction to Robotics 2e, TATA McGraw Hills Education (2014)
2. AsitavaGhoshal, Robotics: Fundamental concepts and analysis, Oxford University Press (2006)
3. Dilip Kumar Pratihar, Fundamentals of Robotics, Narosa Publishing House, (2019)
4. R. K. Mittal, I. J. Nagrath, Robotics and Control, TATA McGraw Hill Publishing Co Ltd, New Delhi (2003)
5. S. B. Niku, Introduction to Robotics – Analysis, Contro, Applications, 3rd edition, John Wiley & Sons Ltd., (2020)
6. J. Angeles, Fundamentals of Robotic Mechanical Systems Theory Methods and Algorithms, Springer (1997)
7. MikellGroover, Mitchell Weiss, Roger N. Nagel, Nicholas Odrey, AshishDutta, Industrial Robotics 2nd edition, SIE, McGraw Hill Education (India) Pvt Ltd (2012)
8. R. D. Klafter, Thomas A. Chmielewski, and MechaelNegin, Robotic Engineering – An Integrated Approach, EEE, Prentice Hall India, Pearson Education Inc. (2009)
9. J. J. Craig, “Introduction to Robotics: Mechanics and Control”, 3rd edition, AddisonWesley (2003)